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# Corrigendum: Bond Risk Premiums with Machine Learning

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In this note we revisit the empirical results in Bianchi, Büchner, and Tamoni (2020) after correcting for using information not available at the time the forecast was made. Although we note a decrease in out-of-sample $R^2$, the revised analysis confirms that bond excess return predictability from neural networks remains statistically and economically significant. (*JEL* )

## Overview

In this note we address an aspect related to the empirical execution in https://doi.org/10.1093/rfs/hhaa062, "Bond Risk Premiums with Machine Learning" by Daniele Bianchi, Matthias Büchner, and Andrea Tamoni. To set the stage, it is important to redefine notation. Specifically, define the 1-year holding period return as $r^{(n)}_{t+12} = \log \frac{P^{(n-12)}_{t+12}}{P^{(n)}_t}$ where the time units are in months (i.e., 12 refers to 12 months). In forecasting $r^{(n)}_{t+12}$, Bianchi, Büchner, and Tamoni (2020) made use of information brought about by $r^{(n)}_{t+11} = \log \frac{P^{(n-12)}_{t+11}}{P^{(n)}_{t-1}}$ (this was

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the case also for the benchmark historical returns).[1] Due to the high persistence in bond returns, it is important to re-evaluate the findings in Bianchi, Büchner, and Tamoni (2020) when one calculates the forecasts of $r_{t+12}^{(n)}$ exploiting the return series only up to $r_t^{(n)}$.

We revisit the main empirical analysis in the original paper. Specifically, we consider both a framework that exploits information in the yield curve only, as in Cochrane and Piazzesi (2005), as well as a framework that uses information from a dataset of a large set of macroeconomic indicators as in Ludvigson and Ng (2009). We employ the same list of models analyzed in the main paper to ensure full comparison of the results.[2]

Moreover, in addition to the analysis of 1-year holding period excess returns, we also report evidence for the 1-month holding period returns. This analysis appeared in an early draft of the paper, but was dropped during the revision process. However, this evidence is important because it provides a cleaner setup to test the effect of overlapping returns on the main results of the paper.

The empirical evidence from the revised analysis confirms that: (1) within each empirical application (i.e., yields-only or yields plus macroeconomic variables) neural networks (NNs) outperform sparse and dense linear models; in particular, NNs are able to detect predictable variations in bond excess returns, as indicated by out-of-sample predictive $R^2$s that, despite being smaller than those in the original paper, are still positive across maturities and statistically significant; (2) using information from macroeconomic and financial variables improves the predictive accuracy of NN forecasts based only on (potentially

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1 Note that defining $r_{t,t+1} = \log \frac{P_t+1}{P_t}$ for stock returns, we have $r^s_{t,t+12} = r_{t,t+1} + r_{t+1,t+2} + r_{t+2,t+3} + r_{t+3,t+4} + r_{t+4,t+5} + r_{t+5,t+6} + r_{t+6,t+7} + r_{t+7,t+8} + r_{t+8,t+9} + r_{t+9,t+10} + r_{t+10,t+11} + r_{t+11,t+12}$, and $r^s_{t-1,t+11} = r_{t-1,t} + r_{t,t+1} + r_{t+1,t+2} + r_{t+2,t+3} + r_{t+3,t+4} + r_{t+4,t+5} + r_{t+5,t+6} + r_{t+6,t+7} + r_{t+7,t+8} + r_{t+8,t+9} + r_{t+9,t+10} + r_{t+10,t+11}$; there are clearly 11 observations that are exactly the same between $r^s_{t-1,t+11}$ and $r^s_{t,t+12}$. However, for bonds this is not immediately the case since the $n$-year bond becomes an $n-1$ month bond, 1 month later. Indeed,

$$r_{t+12}^{(n)} = \log \left( \frac{P_{t+1}^{(n-1)}}{P_t^{(n)}} \frac{P_{t+2}^{(n-2)}}{P_{t+1}^{(n-1)}} \frac{P_{t+3}^{(n-3)}}{P_{t+2}^{(n-2)}} \frac{P_{t+4}^{(n-4)}}{P_{t+3}^{(n-3)}} \frac{P_{t+5}^{(n-5)}}{P_{t+4}^{(n-4)}} \frac{P_{t+6}^{(n-6)}}{P_{t+5}^{(n-5)}} \frac{P_{t+7}^{(n-7)}}{P_{t+6}^{(n-6)}} \frac{P_{t+8}^{(n-8)}}{P_{t+7}^{(n-7)}} \frac{P_{t+9}^{(n-9)}}{P_{t+8}^{(n-8)}} \frac{P_{t+10}^{(n-10)}}{P_{t+9}^{(n-9)}} \frac{P_{t+11}^{(n-11)}}{P_{t+10}^{(n-10)}} \frac{P_{t+12}^{(n-12)}}{P_{t+11}^{(n-11)}} \right)$$

vs.

$$r_{t+11}^{(n)} = \log \left( \frac{P_{t}^{(n-1)}}{P_{t-1}^{(n)}} \frac{P_{t+1}^{(n-2)}}{P_{t}^{(n-1)}} \frac{P_{t+2}^{(n-3)}}{P_{t+1}^{(n-2)}} \frac{P_{t+3}^{(n-4)}}{P_{t+2}^{(n-3)}} \frac{P_{t+4}^{(n-5)}}{P_{t+3}^{(n-4)}} \frac{P_{t+5}^{(n-6)}}{P_{t+4}^{(n-5)}} \frac{P_{t+6}^{(n-7)}}{P_{t+5}^{(n-6)}} \frac{P_{t+7}^{(n-8)}}{P_{t+6}^{(n-7)}} \frac{P_{t+8}^{(n-9)}}{P_{t+7}^{(n-8)}} \frac{P_{t+9}^{(n-10)}}{P_{t+8}^{(n-9)}} \frac{P_{t+10}^{(n-11)}}{P_{t+9}^{(n-10)}} \frac{P_{t+11}^{(n-12)}}{P_{t+10}^{(n-11)}} \right)$$

with no immediate evidence of overlapping observations. The issue arises because of the time-series autocorrelation in bond returns (e.g., for the 10-year bond, the autocorrelation of the annual holding-period return series is ca. 92% when using monthly, overlapping observations).

2 The Python source code to replicate the main results in the paper as well as the Python code to replicate the results of this note is available on the publisher's website.

nonlinear transformations of) the yield curve; (3) the economic gains from NN forecasts based on macroeconomic and yield information are significantly larger than those obtained using yields alone.

### Research Design

We consider as a burn-in sample the period from 1971:08 to 1989:01.[3] Using only information up until the end of this period we estimate the forecasting model for individual bond excess returns and lagged predictors (macro and/or yields). Due to the predictive nature of the regression, the last observation in the right-hand-side variables is that of January 1988. We use the estimates and the value of the predictors on January 1989 to produce out-of-sample forecasts of 1-year excess returns for each maturity. The first forecast error obtains by comparing the excess holding period return during the February 1989 through January 1990 period and its forecast made on January 1989. We then include the February 1989 information and follow the same procedure to produce forecasts of the March 1989 through February 1990 returns, and so on until the end of 2018.[4]

### Bond Return Predictability and the Yield Curve

Table R1 displays results when we forecast excess returns of Treasury bonds with the yield curve. Panel A of Table R1 displays the results for PCRs and partial least squares (PLS). Consistent with the original paper, the predictive $R^2$ are negative across different maturities. A parsimonious representation with only three PCs significantly outperforms the specification with five and ten PCs, particularly at longer maturities.

Panel B displays the results from various configurations of linear penalized regressions. Differently from the paper, we now see that sparse modeling performs poorly out-of-sample, independently from the bond maturity. In fact, they perform even worse than a parsimonious data compression method (PCA) for almost all maturities.

Panel C of Table R1 shows the results for the nonlinear methods. With the only difference of regression tree methods, the main results of the paper holds. That is, neural networks continue to attain good performance with significantly positive $R^2_{oos}$ across maturities. We also observe that a shallow network with a single hidden layer and three nodes performs only marginally worse than the best, deeper network with two hidden layers and three nodes. The $R^2_{oos}$

[3] We use the Liu and Wu (2019) dataset, for which the full set of ten forward rates is available starting from 1971:08.

[4] In the exercise with 1-month holding periods, we use the Center for Research in Security Prices Treasury Fama bond portfolios. In this case the sample period is from 1964:01 to 2016:12, as in the draft version of the paper. The start of the out-of-sample period remains the same, i.e. January 1990.

**Table R1**
**Forecasting nonoverlapping annual holding-period returns with forward rates**

| Models | $R^2_{oos}$ | | | | | | $p$-value | | | | | | $R^2_{oos}$ EW | $p$-value EW |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | $xr^{(2)}_{t+1}$ | $xr^{(3)}_{t+1}$ | $xr^{(4)}_{t+1}$ | $xr^{(5)}_{t+1}$ | $xr^{(7)}_{t+1}$ | $xr^{(10)}_{t+1}$ | $xr^{(2)}_{t+1}$ | $xr^{(3)}_{t+1}$ | $xr^{(4)}_{t+1}$ | $xr^{(5)}_{t+1}$ | $xr^{(7)}_{t+1}$ | $xr^{(10)}_{t+1}$ | $xr^{(EW)}_{t+1}$ | $xr^{(EW)}_{t+1}$ |
| **Panel A:** PCA and PLS | | | | | | | | | | | | | | |
| PCA (10 of 10 components) | −92.8% | −71.0% | −59.5% | −49.9% | −37.4% | −20.8% | | | | | | | −45.2% | |
| PCA (5 of 10 components) | −78.3% | −59.4% | −49.1% | −36.9% | −30.1% | −14.7% | | | | | | | −36.0% | |
| PCA (3 of 10 components) | −30.7% | −22.2% | −15.9% | −8.1% | −5.3% | 3.4% | | | | | | | −7.8% | |
| PCA-squared (5 of 10 components) | −99.3% | −83.8% | −70.6% | −57.1% | −47.0% | −31.1% | | | | | | | −55.7% | |
| PCA-squared (3 of 10 components) | −86.8% | −77.4% | −66.3% | −52.9% | −47.0% | −37.5% | | | | | | | −55.5% | |
| Partial least squares (5 components) | −93.4% | −73.9% | −64.8% | −54.1% | −43.5% | −30.0% | | | | | | | −52.0% | |
| Partial least squares (3 components) | −85.2% | −65.0% | −54.3% | −43.4% | −32.2% | −15.7% | | | | | | | −39.3% | |
| **Panel B:** Penalized linear regressions | | | | | | | | | | | | | | |
| Ridge | −67.0% | −49.3% | −40.8% | −31.4% | −24.5% | −10.8% | | | | | | | −29.5% | |
| LASSO | −22.9% | −23.0% | −20.0% | −19.7% | −21.2% | −15.6% | | | | | | | −19.4% | |
| Elastic net | −22.5% | −22.5% | −24.2% | −18.1% | −23.3% | −18.1% | | | | | | | −21.2% | |
| **Panel C:** Regression trees and neural networks | | | | | | | | | | | | | | |
| Gradient boosted tree | −64.9% | −73.8% | −75.1% | −64.4% | −62.4% | −53.2% | | | | | | | −64.2% | |
| Random forest | −18.4% | −13.7% | −11.7% | −7.4% | −8.2% | −5.0% | | | | | | | −9.0% | |
| Extreme tree | −26.6% | −22.9% | −20.6% | −17.2% | −18.3% | −17.0% | | | | | | | −20.0% | |
| NN - 1 layer (3 nodes) | 2.6% | 4.3% | 4.5% | 4.4% | 4.4% | 3.8% | 16.8% | 10.1% | 9.5% | 8.8% | 10.6% | 12.6% | 4.4% | 10.4% |
| NN - 1 layer (5 nodes) | 2.8% | 4.3% | 4.9% | 4.6% | 4.5% | 3.8% | 15.6% | 8.3% | 6.7% | 5.9% | 7.6% | 9.5% | 4.6% | 7.6% |
| NN - 1 layer (7 nodes) | 2.8% | 4.0% | 4.3% | 3.9% | 3.8% | 3.2% | 12.4% | 6.0% | 5.1% | 5.1% | 7.3% | 10.3% | 3.9% | 6.8% |
| NN - 2 layer (3 nodes each) | 5.5% | 6.5% | 6.8% | 5.9% | 6.2% | 5.1% | 2.8% | 1.1% | 1.0% | 1.0% | 1.4% | 2.6% | 6.2% | 1.4% |
| NN - 2 layer (5 nodes each) | 5.9% | 6.6% | 6.5% | 5.7% | 5.8% | 5.1% | 2.7% | 1.5% | 1.5% | 1.5% | 2.4% | 3.9% | 6.1% | 2.1% |
| NN - 2 layer (7 nodes each) | 6.8% | 6.8% | 6.4% | 5.5% | 5.6% | 4.6% | 1.4% | 1.0% | 1.0% | 1.2% | 1.7% | 3.3% | 5.9% | 1.5% |
| NN - 3 layer (3 nodes each) | 4.0% | 4.5% | 4.3% | 3.9% | 3.2% | 2.3% | 1.1% | 0.5% | 0.3% | 0.4% | 0.2% | 0.7% | 3.6% | 0.3% |
| NN - 3 layer (5 nodes each) | 4.5% | 4.5% | 4.4% | 3.5% | 3.3% | 2.6% | 1.1% | 0.3% | 0.2% | 0.2% | 0.1% | 0.5% | 3.6% | 0.1% |
| NN - 3 layer (7 nodes each) | 5.0% | 4.7% | 4.9% | 4.1% | 3.9% | 3.0% | 0.9% | 0.1% | 0.1% | 0.1% | 0.1% | 0.5% | 4.1% | 0.1% |
| NN - 3 layer (5,4,3 nodes each) | 4.6% | 5.1% | 4.9% | 4.0% | 3.9% | 2.8% | 0.6% | 0.1% | 0.1% | 0.2% | 0.1% | 0.4% | 4.1% | 0.1% |
| Equally weighted model (across all NNs) | 4.9% | 5.4% | 5.5% | 4.8% | 4.7% | 3.9% | 3.4% | 1.4% | 1.1% | 1.1% | 1.8% | 3.4% | 4.9% | 1.6% |
| Inverse val. loss weighted model (across all NNs) | 4.8% | 5.5% | 5.5% | 4.9% | 4.8% | 4.0% | 3.8% | 1.6% | 1.4% | 1.3% | 2.1% | 3.8% | 5.0% | 1.9% |

This table reports the out-of-sample $R^2_{oos}$ obtained using forward rates to predict annual excess bond returns for different maturities and across methodologies. The monthly 1-year holding period returns have been lagged by 11 months in order to mitigate the effect of overlapping returns on the forecasting performance. To compute the out-of-sample $R^2_{oos}$ we compare the forecasts obtained from each methodology to the expectation hypothesis (i.e., to the prediction based on the historical mean). In addition to the $R^2_{oos}$ we report the $p$-value for the null hypothesis $R^2_{oos} \leq 0$ calculated as in Clark and West (2007). Notice that we report a $p$-value only when the $R^2_{oos}$ is positive. The out-of-sample prediction errors are obtained by a recursive forecast which starts in January 1990. The sample period is from 1971:08-2018:12.

for this specification is 6.2% compared to 16.4% in the original paper. Further increasing the depth of the network deteriorates its performance. This continues to be the case even when we consider alternative structures, like a NN with three hidden layers and pyramidal node architecture. In line with the original paper, we conclude that shallow networks perform best.

One may argue that the forecasting results can depend on a particular structure of neural network. To address this issue, we complement and extend some of the results reported in the original paper and report the $R^2_{oos}$ for two model combination strategies that are based on two alternative model averaging schemes: the first model combination scheme combines forecasts of each neural network based on the inverse of the validation loss, which is akin to combining forecasts based on a pseudo-out-of-sample prediction error. The second combination scheme is more agnostic and simply gives equal weights to the forecasts of each neural network. While an equal-weight strategy may seem overly simplistic, the forecasting literature has shown that it may outperform the optimal weights based on log-score or in-sample calibration (see, e.g., Timmermann 2004, Smith and Wallis 2009, and Diebold and Shin 2017). The results show rather unequivocally that even when averaged out, neural networks perform best in out-of-sample forecasting of annual holding period returns.

To summarize, our analysis correcting for nonoverlapping returns confirms that NNs constitute the best-performing methods even in the case when only information in the term structure is used to forecast bond returns (i.e., in a low dimensional setting). This, again, suggests that the gain from nonlinear machine learning methods may not necessarily be relegated to a big data context. Overall, we continue to confirm the main results in Section 3.1 of Bianchi, Büchner, and Tamoni (2020) albeit with lower out-of-sample $R^2$s.

## Bond Return Predictability and Macroeconomic Variables

Next, we consider the setup where information embedded in the yield curve does not necessarily subsume information contained in macro variables. The results in Panels A and B in Table R2 show that: (1) dense modeling, such as data compression techniques and ridge regression, tends to perform poorly out-of-sample; and (2) sparse modeling with both regularization and shrinkage (i.e., elastic net regressions), performs well at long maturities, particularly when restricting the linear combination of forward rates.

Turning to nonlinear machine learning methods in Panel C of Table R2, we observe that the performance of hybrid networks stands out. The only difference relative to the original paper is that increasing the depth of the NN from one- to three-layers does not improve its accuracy.

We also confirm that a careful choice of network structure based on prior economic information exerts a great impact on performance. In particular, a one-layer *group ensembled* model performs better than the *hybrid* NN for the

**Table R2**
**Forecasting nonoverlapping annual holding-period returns with forward rates and macroeconomic variables**

| Models | $R^2_{oos}$ | | | | | | $p$-value | | | | | | $R^2_{oos}$ EW | $p$-value EW |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | $xr^{(2)}_{t+1}$ | $xr^{(3)}_{t+1}$ | $xr^{(4)}_{t+1}$ | $xr^{(5)}_{t+1}$ | $xr^{(7)}_{t+1}$ | $xr^{(10)}_{t+1}$ | $xr^{(2)}_{t+1}$ | $xr^{(3)}_{t+1}$ | $xr^{(4)}_{t+1}$ | $xr^{(5)}_{t+1}$ | $xr^{(7)}_{t+1}$ | $xr^{(10)}_{t+1}$ | $xr^{(EW)}_{t+1}$ | $xr^{(EW)}_{t+1}$ |
| **Panel A:** PCA and PLS | | | | | | | | | | | | | | |
| PCA - first 8 PC'S | −24.3% | −12.8% | −7.8% | −3.8% | −2.7% | −1.8% | | | | | | | −4.7% | |
| PCA as in LN | −9.0% | −4.7% | −3.1% | −2.9% | −6.0% | −10.3% | | | | | | | −5.9% | |
| PLS - 8 components | −105.0% | −69.0% | −55.4% | −46.3% | −33.3% | −20.8% | | | | | | | −40.7% | |
| **Panel B:** Penalized linear regressions | | | | | | | | | | | | | | |
| Ridge (using CP factor) | −122.6% | −82.8% | −64.4% | −51.7% | −35.2% | −18.7% | | | | | | | −41.3% | |
| Lasso (using CP factor) | −31.6% | −16.6% | −10.4% | −9.3% | −5.9% | 6.3% | | | | | | 0.1% | −2.9% | |
| Elastic net (using CP factor) | −33.4% | −14.3% | −8.5% | −4.0% | 5.7% | 13.1% | | | | | 0.2% | 0.0% | 2.4% | 0.3% |
| Ridge (using fwd rates directly) | −146.9% | −111.6% | −99.8% | −87.7% | −73.5% | −52.9% | | | | | | | −79.5% | |
| Lasso (using fwd rates directly) | −22.2% | −15.1% | −8.8% | −7.3% | −1.4% | 1.5% | | | | | | 0.9% | −3.3% | |
| Elastic net (using fwd rates directly) | −27.8% | −15.3% | −10.0% | −4.9% | 2.2% | 5.7% | | | | | 1.1% | 0.4% | −0.6% | |
| **Panel C:** Regression trees and neural networks | | | | | | | | | | | | | | |
| Gradient boosted tree | −17.4% | −14.0% | −9.6% | −6.3% | −7.7% | −11.4% | | | | | | | −7.9% | |
| Random forest | −4.2% | −12.0% | −8.9% | −7.0% | −6.3% | −2.3% | | | | | | | −5.9% | |
| Extreme tree | −3.2% | −1.3% | −1.5% | −0.7% | −3.3% | 2.0% | | | | | | 8.3% | −0.7% | |
| NN 1 layer (32 nodes), fwd rates direct | 3.4% | 3.2% | 8.6% | 16.7% | 21.1% | 28.2% | 0.1% | 0.0% | 0.0% | 0.0% | 0.0% | 0.0% | 19.6% | 0.0% |
| NN 2 layer (32, 16 nodes), fwd rates direct | −37.5% | −6.2% | −5.5% | 1.5% | 6.9% | 14.5% | | | | 0.0% | 0.0% | 0.0% | 3.6% | 0.0% |
| NN 3 layer (32, 16, 8 nodes), fwd rates direct | −5.7% | 2.4% | 5.5% | 9.0% | 9.5% | 13.8% | | 0.5% | 0.3% | 0.2% | 0.1% | 0.1% | 8.8% | 0.2% |
| NN 1 layer (32 nodes), fwd rates net (2 layer: 3, 3 nodes) | 0.1% | 15.0% | 15.5% | 16.7% | 16.0% | 22.1% | 1.2% | 0.3% | 0.2% | 0.1% | 0.1% | 0.0% | 18.2% | 0.1% |
| NN 2 layer (32, 16, nodes), fwd rates net (2 layer: 3, 3 nodes) | −9.3% | 1.9% | 2.1% | 4.0% | 7.0% | 11.5% | | 0.2% | 0.1% | 0.1% | 0.0% | 0.0% | 6.5% | 0.1% |
| NN 3 layer (32, 16, 8 nodes), fwd rates net (2 layer: 3, 3 nodes) | −13.4% | −1.2% | −1.2% | 1.8% | 2.3% | 6.1% | | | | 0.3% | 0.1% | 0.1% | 2.2% | 0.2% |
| NN 1 layer group ensem (1 node per group), fwd rates direct | −9.2% | 2.4% | 8.1% | 12.5% | 14.1% | 16.4% | | 0.6% | 0.2% | 0.1% | 0.0% | 0.0% | 12.1% | 0.0% |
| NN 2 layer group ensem (2, 1 nodes per group / hidden layer), fwd rates direct | −2.1% | 6.2% | 9.5% | 12.2% | 13.8% | 16.7% | | 2.1% | 0.9% | 0.3% | 0.1% | 0.0% | 12.6% | 0.2% |
| NN 1 layer group ensem (1 node per group), fwd rates net (2 layer: 3, 3 nodes) | 3.6% | 8.9% | 11.3% | 10.9% | 12.9% | 17.0% | 0.9% | 0.4% | 0.3% | 0.1% | 0.1% | 0.0% | 13.3% | 0.1% |
| NN 2 layer group ensem (2, 1 node per group), fwd rates net (2 layer: 3, 3 nodes) | −9.5% | 2.1% | 1.7% | 5.8% | 8.4% | 13.7% | | 1.5% | 1.2% | 0.6% | 0.3% | 0.1% | 8.0% | 0.3% |
| Equally weighted model (across all NNs) | 7.1% | 15.1% | 16.2% | 18.4% | 20.1% | 24.0% | 0.1% | 0.0% | 0.0% | 0.0% | 0.0% | 0.0% | 20.3% | 0.0% |
| Inverse val. loss weighted model (across all NNs) | 6.0% | 14.0% | 15.2% | 17.4% | 19.3% | 23.3% | 0.1% | 0.0% | 0.0% | 0.0% | 0.0% | 0.0% | 19.4% | 0.0% |

This table reports the out-of-sample $R^2_{oos}$ obtained using forward rates and a large panel of macroeconomic variables to predict annual excess bond returns for different maturities. The monthly 1-year holding period returns have been lagged by 11 months in order to mitigate the effect of overlapping returns on the forecasting performance. To compute the out-of-sample $R^2_{oos}$ we compare the forecasts obtained from each methodology to the expectation hypothesis (i.e., prediction based on the historical mean). In addition to the $R^2_{oos}$ we report the $p$-value for the null hypothesis $R^2_{oos} \leq 0$ calculated as in Clark and West (2007). Notice that we report a $p$-value only when the $R^2_{oos}$ is positive. The out-of-sample prediction errors are obtained by a recursive forecast which starts in January 1990. The sample period is from 1971:08-2018:12. Penalized regressions are estimated including macro-economic variables plus either raw forward rates or a linear combination of forward rates as introduced by Cochrane and Piazzesi (2005) (CP). Similarly, neural networks are estimated either by adding the forward rates as additional regressors in the output layer ("fwd rates direct") or by estimating a separate network for forward rates and ensembling both macro and forward rates networks in the output layer ("fwd rates net").

2-, 3-, 4-years maturities and attains slightly lower predictive accuracy for long-term bonds. Also for group ensembling NN, adding more layers is detrimental to the performance.

In line with the analysis reported for the case with yields only above, we also compute the performance of model combination strategies that are based on a weighted average of each neural network's forecasts (see also Section 4.5 of Bianchi, Büchner, and Tamoni (2020)). This is a natural strategy to exploit, since we have observed that different network structures perform differently at short- and long-maturity (e.g., the 1-layer NN with group ensembling and the hybrid 1-layer network perform best at short- and long-maturity, respectively). As before, we consider a forecast combination strategy whereby each neural network is weighted based on the inverse of the validation loss, as well as a second, more agnostic, combination scheme which gives equal weight to each neural network prediction. The results show that as a class of model, on average, neural networks unequivocally outperform both data compression, penalized regression and shallow nonlinear models, therefore confirming the main results of the original paper. The out-of-sample $R^2$ for the best NN model (among those considered in the original paper) is 19.6%. That same specification had an $R^2_{oos}$ of 22.4% in the original paper.

Finally, comparing Table R1 to Table R2, we confirm that macroeconomic variables carry information that is not contained in the yield curve. For instance, a group ensembled NN that exploits macroeconomic and term structure information attains out-of-sample $R^2$s that are about twice as large as the best-performing NN that employs yields only for maturities ranging from 2 to 10 years.

### Economic Value of Return Forecasts from Neural Networks

We re-examine the findings in Section 5 of Bianchi, Büchner, and Tamoni (2020). Again, since our goal is solely to correct for the persistence in 1-year holding period returns, we consider an asset allocation framework that is identical to that described in Section 5.1 of the main paper. Table R3 shows the annualized certainty equivalent return (CER) values computed relative to the EH model. Positive values indicate that the predictive model performs better than the EH model. For the sake of completeness, we focus on the predictions implied by (1) the best performing neural network as in Tables R1-R2, (2) a model-average neural-network based on equal weights, (3) a model-average neural network where the weights of each model are based on the validation loss, and (4) for the exercise with macro + fwd rates the group-ensemble approach outlined in the main paper.[5] We flag in bold those values that are statistically significant at the 5% confidence level.

[5] The economic analysis of model combination forecasts did not appear in the published manuscript and is new to this corrigendum.

**Table R3**
**Economic significance of forecasts for nonoverlapping annual returns**

Panel A: Mean-variance utility

| | 2-year maturity | | | 10-year maturity | | | All | | |
|---|---|---|---|---|---|---|---|---|---|
| | Fwd rates | Fwd + macro | Δ | Fwd rates | Fwd + macro | Δ | Fwd rates | Fwd + macro | Δ |
| Best performing NN | 0.29 | 0.30 | 0.01 | **0.77** | **6.20** | **5.44** | 1.08 | **3.45** | **2.37** |
| p-value | (0.085) | (0.108) | (0.901) | (0.044) | (0.000) | (0.000) | (0.377) | (0.038) | (0.045) |
| Group-ensemble NN | | 0.30 | 0.01 | | **4.48** | **3.71** | | 3.03 | 1.95 |
| p-value | | (0.077) | (0.191) | | (0.000) | (0.000) | | (0.114) | (0.075) |
| NN model averaged EW | 0.30 | **0.40** | 0.11 | **0.66** | **5.54** | **4.88** | 0.04 | **2.06** | **2.02** |
| p-value | (0.078) | (0.024) | (0.202) | (0.004) | (0.000) | (0.000) | (0.150) | (0.000) | (0.002) |
| NN model averaged VL | 0.30 | **0.41** | 0.11 | **0.67** | **5.43** | **4.76** | 0.03 | **1.92** | **1.89** |
| p-value | (0.078) | (0.022) | (0.210) | (0.004) | (0.000) | (0.000) | (0.130) | (0.000) | (0.001) |

Panel B: Power utility

| | 2-year maturity | | | 10-year maturity | | | All | | |
|---|---|---|---|---|---|---|---|---|---|
| | Fwd rates | Fwd + macro | Δ | Fwd rates | Fwd + macro | Δ | Fwd rates | Fwd + macro | Δ |
| Best performing NN | 0.26 | 0.27 | 0.01 | **0.78** | **6.14** | **5.35** | 1.25 | **3.81** | **2.56** |
| p-value | (0.103) | (0.135) | (0.454) | (0.046) | (0.000) | (0.000) | (0.209) | (0.031) | (0.032) |
| Group-ensemble NN | | 0.27 | 0.01 | | **4.46** | **3.68** | | 3.33 | 1.98 |
| p-value | | (0.096) | (0.191) | | (0.000) | (0.000) | | (0.088) | (0.065) |
| NN model averaged EW | 0.27 | **0.37** | 0.10 | **0.67** | **5.49** | **4.82** | 0.17 | **2.22** | **2.05** |
| p-value | (0.093) | (0.030) | (0.230) | (0.004) | (0.000) | (0.000) | (0.175) | (0.000) | (0.003) |
| NN model averaged VL | 0.27 | **0.37** | 0.11 | **0.69** | **5.36** | **4.69** | 0.16 | **2.05** | **1.88** |
| p-value | (0.098) | (0.028) | (0.210) | (0.005) | (0.000) | (0.000) | (0.168) | (0.000) | (0.001) |

This table reports the annualized certainty equivalent values (in %) for portfolio decisions based on the out-of-sample forecasts of bond excess returns for an investor with either mean-variance (Panel A) or power utility (Panel B) and a coefficient of risk aversion equal to five. The table reports two asset allocation exercises. In the univariate asset allocation case, the investor selects either the 2- or the 10-year bond, along with the 1-year short-rate. In the multivariate case, the investor selects bonds across the six maturities, 2- to 5-, 7- and 10-years. The asset allocation decision is based on the predictions implied by (1) the best performing neural network as in Tables R1-R2, (2) a model-average neural-network based on equal weights, (3) a model-average neural network where the weights of each model are based on the validation loss, and (4) for the exercise with macro + fwd rates the group-ensemble approach outlined in the main paper. The models are benchmarked against the expectation hypothesis. We also test for the difference in each model performance between a forecast using only fwd rates (CP) and both fwd rates and macroeconomic variables (LN). The out-of-sample predictions are obtained by a recursive forecast which starts in January 1990. The sample period is from 1971:08-2018:12. Statistical significance is based on a one-sided Diebold and Mariano (1995) test as extended by Harvey, Leybourne, and Newbold (1997) to account for autocorrelation in the forecasting errors. We flag in bold those values that are statistically significant at the 5% confidence level.

Two facts emerge; first, there is strong evidence for the economic value of using neural network forecasts both for the univariate case exploiting the long maturity bond, and for the multivariate asset allocation strategy. Second, including macroeconomic information in the conditioning information set significantly increases the economic performance of the forecasts; specifically, the NN forecasts that exploit macroeconomic information produce significantly higher CER values than those implied by a NN using yields-only (for the univariate and multivariate allocation, independently of the form of utility).

Overall, with now the exception of regression trees, we confirm the analysis in Section 5.2 of Bianchi, Büchner, and Tamoni (2020) that neural network-based forecasts of bond returns provide support for the hypothesis that a (statistically and economically) significant portion of macroeconomic information is not captured by the yield curve, even after accounting for nonlinearity in interest rates.

## Monthly Holding-Period Returns

As we have already stressed, the goal of this corrigendum is to investigate the robustness of the empirical results reported in the published version when correcting for the overlapping observations induced by the 1-year holding period returns.

In this section, however, we also report some additional results which are not included in the original paper. In particular, we report the forecasting performance of the full battery of linear and nonlinear models for the 1-month monthly holding period returns. Arguably, this setting is ideal to investigate the effect of overlapping returns on the performance of neural networks.

As a matter of fact, Bollerslev, Tauchen, and Zhou (2009) show that short-horizon stock returns are very hard to predict. Similarly, the monthly predictability of bond excess returns tends to be much lower than predictability of annual holding-period-returns (see, e.g., Gargano, Pettenuzzo, and Timmermann 2019).

Tables R4 and R5 report the results for the sample period from 1964:01 to 2016:12, for the yield-only and the yield + macro exercises, respectively.[6] When forecasting bond returns based on information from the term structure only, we observe that the ranking displayed in Table R1 is largely preserved. That is, the performance (averaged across maturities) of nonlinear models, specifically NNs, is higher than the one obtained by data compression methods and penalized linear regressions. For instance, at longer maturities neural networks attain an $R^2_{oos}$ which is up to 50% bigger than the parsimonious PCA with three factors. The evidence in Tables R1 and R4 show that NN models

[6] The test assets are the Center for Research in Security Prices Treasury Fama bond portfolios. These very same assets have been used, e.g., by Duffee (2002). Only non callable, non flower notes and bonds are included in the portfolios. The portfolio returns are an equal-weighted average of the unadjusted holding period return for each bond in the portfolios in excess of the risk-free rate. We consider four buckets: bonds with maturity of 1-2 years, 2-3 years, 3-4 years, and 4-5 years.

are able to detect substantial bond return predictability with $R^2_{oos}$s that are in positive territory for both monthly and annual holding period returns.

Turning our attention to the forecasting exercise that exploits both yields and macroeconomic variables, Table R5 largely confirms the results obtained for annual bond excess returns. In particular, Panel C shows that imposing a sensible economic structure through group ensembling significantly increases the out-of-sample predictive performance of the (two-layer hybrid) NNs. That is, grouping variables within economic categories and then training a shallow network within each group – a network structure that we dub "group ensembling"– attains a performance that is better than the best-performing deep NN where no economic priors are utilized. Thus, the depth of the network and the economic priors used to design the network (e.g., grouping within categories) interact with one another, a result that has been highlighted in Bianchi, Büchner, and Tamoni (2020) and that is new to the empirical finance literature. Differently from Table R2, we also see that extreme trees perform competitively to neural networks when using monthly holding period returns.

Further, comparing Table R4 to Table R5, we find that a group ensemble NN that exploits macroeconomic and term structure information attains out-of-sample $R^2$s that are more than three times as large as the best-performing NN that employs yields only. This result is on par with, if not larger than, the result obtained using annual holding period returns.

Taking stock of the results in Tables R2-R5, we confirm the evidence in Bianchi, Büchner, and Tamoni (2020) that (1) macroeconomic data convey independently relevant information for bond return predictability, and (2) an economically-driven choice of the network structure may perform on par with more data-driven estimates.

## Additional results

In an un-tabulated set of results, we continue to find that bond risk premia obtained from NNs display a clear countercyclical pattern (as discussed in Bianchi, Büchner, and Tamoni (2020), Section 6.1); in particular, the contemporaneous correlations between forecasts of the 10-years excess bond returns and industrial production is -10.9% (p-value of 0.19) when only information in the term structure is used, and this correlation more than doubles to -44.6% (p-value of 0.005) when we add macroeconomic variables to forward rates.

We also verify that the various NN models perform well relative to the EH model throughout the sample period as manifested by cumulative sum of squared errors lines that are positive and increasing for most periods (see Figure R1).

Finally, across all bond maturities and model specifications, (1) the correlation between our forecasts and the subjective bond risk premia of Buraschi, Piatti, and Whelan (2019) is small, and not statistically significant; (2) there is a strong and positive association between our forecasts and the

**Table R4**
**Forecasting monthly holding-period returns with forward rates**

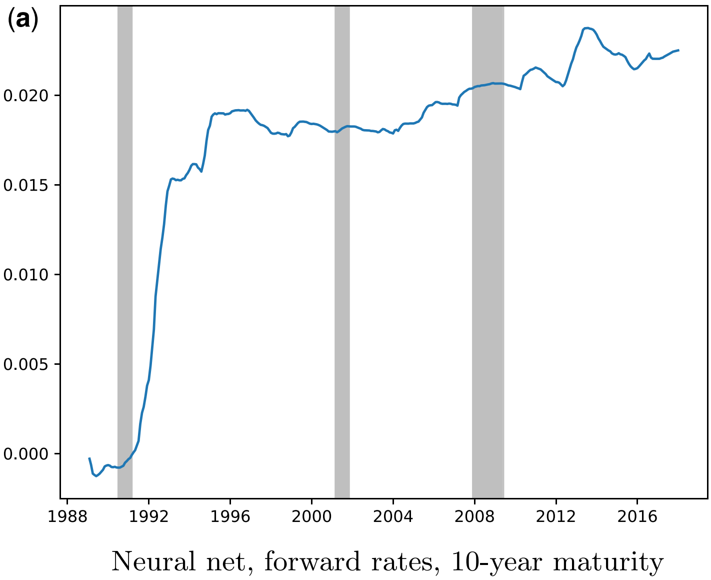
| Models | $R^2_{oos}$ | | | | $p$-value | | | | $R^2_{oos}$ EW | $p$-value EW |
|---|---|---|---|---|---|---|---|---|---|---|
| Maturity (months) | 13-24 | 25-36 | 37-48 | 49-60 | 13-24 | 25-36 | 37-48 | 49-60 | $xr^{(EW)}_{t+1}$ | $xr^{(EW)}_{t+1}$ |
| **Panel A:** PCA and PLS | | | | | | | | | | |
| PCA (5 of 5 components) | −5.3% | −3.7% | −1.3% | −0.3% | | | | | −1.8% | |
| PCA (3 of 5 components) | −0.2% | 0.2% | 1.0% | 1.0% | | 11.3% | 5.8% | 5.1% | 0.7% | 7.1% |
| PCA (1 of 5 components) | −4.4% | −3.3% | −2.8% | −2.5% | | | | | −3.0% | |
| PCA-squared (5 of 5 components) | −7.7% | −4.0% | −1.6% | −0.5% | | | | | −2.1% | |
| PCA-squared (3 of 5 components) | −3.9% | −2.0% | −0.8% | 0.0% | | | | 4.8% | −0.9% | |
| Partial least squares (5 components) | −5.3% | −3.7% | −1.3% | −0.3% | | | | | −1.8% | |
| Partial least squares (3 components) | −4.0% | −2.7% | 0.3% | 0.6% | | | 10.5% | 7.9% | −0.5% | |
| **Panel B:** Penalized linear regressions | | | | | | | | | | |
| Ridge | −1.9% | −1.1% | 0.0% | 0.3% | | | | 12.6% | −0.3% | |
| LASSO | −1.1% | −1.3% | −0.9% | −0.5% | | | | | −0.6% | |
| Elastic net | 0.4% | −0.5% | −0.3% | −0.2% | 24.5% | | | | −0.2% | |
| **Panel C:** Regression trees and neural networks | | | | | | | | | | |
| Gradient boosted tree | −0.2% | −1.0% | −1.0% | −0.3% | | | | | −0.5% | |
| Random forest | −15.9% | −17.0% | −16.0% | −14.7% | | | | | −14.8% | |
| Extreme tree | −16.3% | −24.5% | −27.3% | −21.9% | | | | | −22.8% | |
| NN - 1 layer (3 nodes) | 0.8% | 1.0% | 1.1% | 1.1% | 9.3% | 5.1% | 2.1% | 1.1% | 1.1% | 2.3% |
| NN - 1 layer (5 nodes) | 0.7% | 1.0% | 1.1% | 1.1% | 12.1% | 4.5% | 2.3% | 1.0% | 1.1% | 2.4% |
| NN - 2 layer (3 nodes each) | 0.7% | 1.1% | 1.2% | 1.2% | 10.3% | 4.3% | 2.0% | 1.1% | 1.2% | 2.2% |
| NN - 2 layer (5 nodes each) | 0.8% | 1.3% | 1.4% | 1.5% | 10.1% | 3.5% | 1.2% | 0.5% | 1.4% | 1.5% |
| NN - 3 layer (3 nodes each) | 0.9% | 1.1% | 1.2% | 1.2% | 8.9% | 5.2% | 2.5% | 1.2% | 1.2% | 2.6% |
| NN - 3 layer (5 nodes each) | 0.7% | 1.2% | 1.3% | 1.3% | 11.9% | 4.5% | 2.1% | 0.9% | 1.3% | 2.3% |
| NN - 3 layer (5, 4, 3 nodes each) | 0.7% | 1.1% | 1.3% | 1.3% | 11.7% | 5.1% | 2.1% | 1.0% | 1.2% | 2.4% |

This table reports the out-of-sample $R^2_{oos}$ obtained using forward rates to predict monthly excess bond returns for different maturities and across methodologies. To compute the out-of-sample $R^2_{oos}$ we compare the forecasts obtained from each methodology to the expectation hypothesis (i.e., to the prediction based on the historical mean). In addition to the $R^2_{oos}$ we report the $p$-value for the null hypothesis $R^2_{oos} \leq 0$ calculated as in Clark and West (2007). Notice that we report a $p$-value only when the $R^2_{oos}$ is positive. The out-of-sample prediction errors are obtained by a recursive forecast which starts in January 1990. The sample period is from 1964:01-2016:12.

**Table R5**
**Forecasting monthly holding-period returns with forward rates and macroeconomic variables**

| Models | $R^2_{oos}$ | | | | $p$-value | | | | $R^2_{oos}$ EW | $p$-value EW |
|---|---|---|---|---|---|---|---|---|---|---|
| Maturity (months) | 13-24 | 25-36 | 37-48 | 49-60 | 13-24 | 25-36 | 37-48 | 49-60 | $xr^{(EW)}_{t+1}$ | $xr^{(EW)}_{t+1}$ |
| **Panel A:** PCA and PLS | | | | | | | | | | |
| PCA - first 8 PCs | −12.1% | −4.2% | −0.3% | 3.5% | | | | 0.3% | −0.4% | |
| PCA as in LN | −4.9% | −1.1% | 0.3% | 1.9% | | | 0.5% | 0.8% | 0.3% | 0.6% |
| PLS - 8 components | −68.8% | −54.5% | −39.6% | −30.8% | | | | | −40.6% | |
| **Panel B:** Penalized linear regressions | | | | | | | | | | |
| Ridge (using CP factor) | −61.2% | −49.1% | −35.5% | −27.3% | | | | | −36.5% | |
| Lasso (using CP factor) | 4.2% | 3.2% | 2.6% | 2.6% | 0.5% | 0.3% | 0.3% | 0.4% | 3.2% | 0.4% |
| Elastic net (using CP factor) | 4.6% | 3.5% | 2.0% | 2.5% | 0.4% | 0.3% | 0.4% | 0.3% | 3.1% | 0.4% |
| Ridge (using fwd rates directly) | −61.8% | −49.8% | −36.0% | −27.8% | | | | | −37.1% | |
| Lasso (using fwd rates directly) | 5.1% | 2.9% | 1.1% | 2.0% | 0.2% | 0.5% | 0.6% | 0.4% | 2.6% | 0.4% |
| Elastic net (using fwd rates directly) | 4.8% | 3.5% | 2.0% | 2.3% | 0.3% | 0.3% | 0.5% | 0.5% | 3.0% | 0.4% |
| **Panel C:** Regression trees and neural networks | | | | | | | | | | |
| Gradient boosted regression tree | −1.8% | −3.4% | −2.8% | −5.8% | | | | | −2.1% | |
| Random forest | 3.9% | 1.2% | 2.4% | 1.3% | 0.5% | 0.5% | 0.2% | 3.4% | 2.8% | 0.9% |
| Extreme tree | 6.2% | 3.7% | 2.3% | 3.6% | 0.1% | 0.1% | 0.7% | 0.2% | 4.1% | 0.2% |
| NN 1 layer (32 nodes), fwd rates direct | −1.5% | 0.4% | 2.0% | 4.3% | | 0.9% | 0.7% | 2.2% | 2.6% | 1.3% |
| NN 2 layer (32, 16 nodes), fwd rates direct | 3.6% | 3.4% | 3.8% | 4.7% | 1.0% | 0.7% | 0.6% | 1.7% | 4.3% | 1.0% |
| NN 3 layer (32, 16, 8 nodes), fwd rates direct | −2.8% | −2.1% | −1.8% | −1.6% | | | | | −2.0% | |
| NN 1 layer (32 nodes), fwd rates net (1 layer: 3 nodes) | −1.0% | −0.6% | −0.8% | 0.2% | | | | 6.2% | −0.3% | |
| NN 2 layer (32, 16, nodes), fwd rates net (1 layer: 3 nodes) | 0.6% | 1.0% | 0.9% | 1.9% | 1.4% | 0.7% | 0.6% | 0.7% | 1.4% | 0.5% |
| NN 3 layer (32, 16, 8 nodes), fwd rates net (1 layer: 3 nodes) | 0.8% | 1.1% | 1.2% | 1.2% | 8.0% | 5.2% | 3.4% | 2.4% | 1.2% | 3.3% |
| NN 1 layer group ensem (1 node per group), fwd rates direct | 6.1% | 5.1% | 4.6% | 4.6% | 0.1% | 0.1% | 0.1% | 0.5% | 5.1% | 0.2% |
| NN 2 layer group ensem (2, 1 nodes per group/hidden layer), fwd rates direct | 1.4% | 0.8% | 0.7% | 0.8% | 6.3% | 9.9% | 10.5% | 9.2% | 0.9% | 8.9% |
| NN 1 layer group ensem (1 node per group), fwd rate net (1 layer: 3 nodes) | 5.9% | 5.2% | 5.0% | 5.1% | 0.1% | 0.1% | 0.1% | 0.3% | 5.5% | 0.2% |
| NN 2 layer group ensem (2, 1 node per group), fwd rate net (1 layer: 3 nodes) | 1.5% | 1.3% | 1.2% | 1.1% | 3.9% | 3.6% | 3.9% | 3.9% | 1.3% | 3.4% |

This table reports the out-of-sample $R^2_{oos}$ obtained using forward rates and a large panel of macroeconomic variables to predict monthly excess bond returns for different maturities. To compute the out-of-sample $R^2_{oos}$ we compare the forecasts obtained from each methodology to the expectation hypothesis (i.e., prediction based on the historical mean). In addition to the $R^2_{oos}$ we report the $p$-value for the null hypothesis $R^2_{oos} \leq 0$ calculated as in Clark and West (2007). Notice that we report a $p$-value only when the $R^2_{oos}$ is positive. The out-of-sample prediction errors are obtained by a recursive forecast which starts in January 1990. The sample period is from 1964:01-2016:12. Penalized regressions are estimated including macro-economic variables plus either raw forward rates or a linear combination of forward rates as introduced by Cochrane and Piazzesi (2005) (CP). Similarly, neural networks are estimated either adding the forward rates as additional regressors in the output layer ("fwd rates direct") or by estimating a separate network for forward rates and ensembling both macro and forward rates networks in the output layer ("fwd rates net").



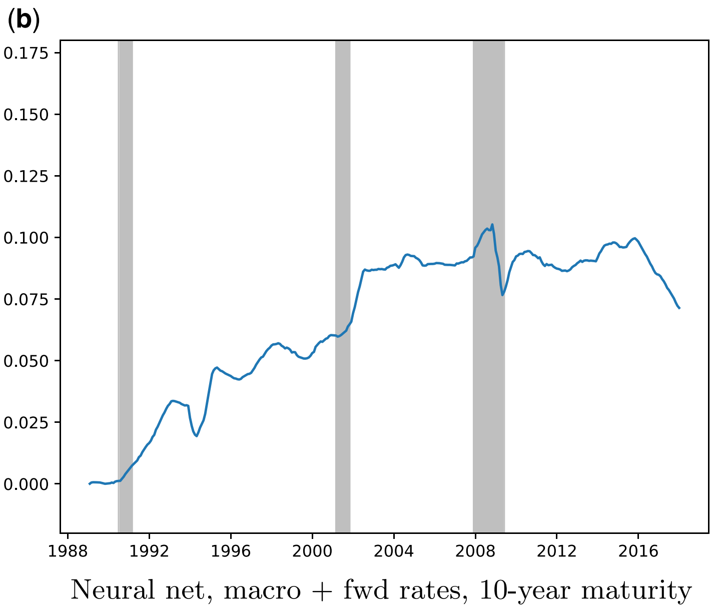




**Figure R1**
**Squared forecast errors across time**
This figure plots the time series of difference in squared forecast errors from a given model versus the expectations hypothesis. We scale the forecast errors by the variance of the dependent variable times $T-1$, i.e. each month $t$ we plot the value attained by $\frac{(\hat{\epsilon}^{EH}_{t+1})^2-(\hat{\epsilon}^{Model}_{t+1})}{(T-1)\mathrm{Var}(rx_{t+1})}$. The out-of-sample period starts in 1990. The expectation hypothesis uses all data starting from the in-sample period in 1971:08. The figures present results for the 10-year maturity and focus on the neural network with two layers and three nodes (NN - 2 layer [3 nodes]), when forecasts are generated based on forward rates only and on the group-ensembled network (NN 1 layer group ensem + fwd rate net) when forecasts are generated based on macroeconomic variables + forward rates.

Piazzesi, Salomao, and Schneider (2015) measure of bond risk premia when yields-only based forecasts are considered; however, we note that adding macroeconomic variables weakens the correlation between our forecasts and subjective measures based on consensus; (3) the correlation between our forecasts and those of Giacoletti, Laursen, and Singleton (2016) are large and mostly significant. These results confirm the analysis in Section 6.3 of Bianchi, Büchner, and Tamoni (2020).

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